

Akshata Madavi

+1-408-590-1556 | akshatamadavi@gmail.com | linkedin.com/in/akshata-madavi | github.com/Akshata4 | San Jose, CA

PROFESSIONAL SUMMARY

MS Software Engineering student at SJSU (GPA 4.0) with 6 years of production data engineering and hands-on deep learning shipped in grad school. Trained and deployed a real-time anomaly detection model (XGBoost, 98% accuracy) on time-series sensor data with end-to-end Kafka streaming inference in ~10 seconds on GCP. Fine-tuned LLMs via LoRA (+10 pp accuracy) and built production ML pipelines processing 10M+ records. Brings the rare combination of large-scale data infrastructure depth and modern AI/ML engineering.

PROJECTS

VitalWatch — Real-Time Sensor Anomaly Detection AI-Catalyst Hackathon 2025

PyTorch, XGBoost, Kafka/KSQL, GCP Vertex AI, BigQuery, Twilio, Python

- Trained an XGBoost model on multi-channel time-series sensor data (heart rate, SpO2, temperature) achieving **98% anomaly detection accuracy**; deployed as a real-time scoring service on GCP Vertex AI.
- Built a Kafka/KSQL streaming inference pipeline; enriched flagged events via Vertex AI LLM with contextual BigQuery data; delivered end-to-end alerts in **~10 seconds**.

Enterprise AI Governance Platform — LLM Fine-tuning & Monitoring

PyTorch, HuggingFace, LoRA, LangChain, FAISS, MLflow, AWS, Docker

- Fine-tuned Qwen 0.6B via LoRA on a custom labeled dataset; tracked experiments with MLflow, improving detection accuracy from **87% to 97% (+10 pp)**.
- Designed vector store retrieval pipelines (LangChain + FAISS) for failure pattern analysis; containerized full stack with Docker for reproducible training and deployment.

WORK EXPERIENCE

LTIMindtree — Senior Data Engineer May 2023 – Mar 2025 | Pune, India

- Engineered near real-time GCP/BigQuery data pipelines ingesting **~10M records**, building ML-ready feature tables for downstream AI/ML models at production scale.
- Redesigned a multi-source ETL service (~30 GB) using parallel Spark jobs, cutting ingestion from **1 week to under 10 minutes**; served as primary feature store for ML workloads.

STL (Sterlite Technologies) — Data Engineer Jun 2018 – May 2021 | Pune, India

- Built near real-time GCP pipelines migrating Oracle, SAP, and Salesforce to BigQuery (**~1 TB**); applied DOE-based statistical analysis for manufacturing scrap detection and process optimization.
- Developed automated anomaly detection checks over sensor and telemetry outputs, reducing manual review time and catching production anomalies earlier.

AI Adventures — Data Engineer Aug 2021 – Apr 2023 | Pune, India

- Built data migration service and BI dashboards; A/B testing and retention analysis of curriculum content drove a **20% revenue increase**.

EDUCATION

MS in Software Engineering — San Jose State University Expected May 2027

GPA: 4.0/4.0 | Machine Learning, Deep Learning, Recommender Systems, Software Systems Engineering

BS in Computer Engineering — College of Engineering Pune (COEP) May 2018

SKILLS

Languages: Python, SQL

ML Frameworks: PyTorch, TensorFlow, scikit-learn, XGBoost, PySpark ML, CNNs

Deep Learning & GenAI: Fine-tuning/LoRA (Qwen, LLaMA), RAG (LangChain, LlamaIndex), Vector stores (FAISS, Pinecone), Evaluation (Ragas, Promptfoo), Serving (Vertex AI, HuggingFace)

Data & Streaming: Kafka/KSQL, Spark, BigQuery, Airflow, Pandas, NumPy, MySQL

MLOps & Cloud: MLflow, Docker, GitHub Actions CI/CD, GCP (Vertex AI, BigQuery, GCS), AWS